

ZHAOTONG, China

WMO: 565860

Lat: 27.3517N

Lon: 103.7189E

Elev: 6398

StdP: 11.60

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.3	26.1	15.5	15.3	47.0	18.5	17.7	43.0	16.6	54.1	14.3	54.1	3.3	0	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	83.6	63.3	81.5	63.3	79.5	62.9	67.5	77.2	66.7	76.1	65.9	74.9	6.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	117.7	71.0	64.1	114.2	70.2	63.3	111.1	69.7	36.3	77.6	35.5	76.6	34.8	75.2	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.4	12.2	10.6	DB	21.2	87.6	2.6	1.7	19.3	88.8	17.8	89.9	16.3	90.8	14.4	92.1	(4)
			WB	19.8	69.6	2.7	1.4	17.9	70.6	16.3	71.4	14.8	72.2	12.9	73.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	38.9	43.4	50.6	58.4	63.0	66.1	69.4	68.6	63.8	56.4	49.4	40.9	
	DBStd	12.27	7.22	9.26	8.11	7.32	6.09	5.00	3.71	3.91	5.74	6.10	6.49	6.20	
	HDD50	1074	349	221	95	18	1	0	0	0	0	14	89	287	
	HDD65	3837	811	606	447	214	112	47	8	14	90	272	469	747	
	CDD50	3189	4	35	115	270	403	482	600	576	416	213	70	5	
	CDD65	476	0	0	2	17	49	79	143	125	55	6	0	0	
	CDH74	2483	0	3	46	207	366	392	612	544	280	33	0	0	
	CDH80	351	0	0	4	23	68	68	81	71	36	0	0	0	
Wind	WSAvg	3.9	3.9	4.8	5.2	5.0	4.3	3.5	3.3	3.2	3.3	3.3	3.3	3.4	
Precipitation	PrecAvg	28.5	0.2	0.2	0.5	1.3	3.2	5.5	5.9	4.6	4.1	2.2	0.7	0.2	
	PrecMax	35.2	0.8	0.8	1.4	3.0	11.0	9.8	13.4	10.0	7.0	6.2	2.7	1.1	
	PrecMin	18.3	0.0	0.0	0.0	0.2	0.2	2.0	2.5	0.9	1.6	0.4	0.0	0.0	
	PrecStd	4.1	0.2	0.2	0.4	0.8	2.2	2.2	2.3	2.0	1.4	1.2	0.6	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.7	73.4	78.9	83.1	85.6	85.4	85.0	84.8	83.9	77.6	72.0	65.5	
		MCWB	45.5	47.8	50.8	55.7	59.2	64.5	66.3	65.4	64.1	60.5	52.5	46.3	
	2%	DB	62.4	69.5	74.3	79.3	81.9	82.0	82.2	82.0	80.1	74.2	68.3	61.1	
		MCWB	43.4	46.2	50.1	55.2	59.8	64.4	65.9	65.3	63.4	58.8	52.1	45.2	
	5%	DB	58.1	65.4	70.8	76.2	78.3	78.6	80.0	79.6	77.3	70.8	64.9	56.5	
		MCWB	42.6	44.9	49.2	54.5	59.5	63.9	65.5	65.1	62.8	57.7	51.0	44.2	
	10%	DB	52.7	61.0	66.2	72.1	74.6	75.4	77.6	77.1	74.2	66.7	60.9	51.4	
		MCWB	41.1	43.8	48.8	54.2	59.0	63.6	65.1	64.7	62.0	56.7	50.1	42.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.3	49.5	54.4	60.4	65.2	68.2	69.0	68.8	66.6	62.8	56.9	48.9
			MCDB	62.8	68.0	69.3	72.0	76.5	78.1	79.2	78.9	76.5	73.6	65.9	61.7
2%		WB	44.8	47.5	52.4	58.4	63.3	67.0	67.7	67.3	65.4	61.0	53.9	46.4	
		MCDB	59.8	64.8	67.9	71.4	74.8	76.5	77.2	77.1	75.2	70.8	64.7	57.9	
5%		WB	43.0	45.9	50.9	57.0	61.9	65.8	66.9	66.4	64.4	59.3	52.1	44.6	
		MCDB	56.5	62.6	65.8	69.7	72.6	74.8	76.0	75.8	73.6	67.4	61.6	55.3	
10%		WB	41.0	44.4	49.4	55.5	60.4	64.7	66.1	65.5	63.3	57.4	50.5	42.6	
		MCDB	52.7	59.8	63.1	68.0	70.3	72.7	74.8	74.1	71.8	64.7	59.0	51.1	
Mean Daily Temperature Range	5% DB	MDBR	18.0	19.3	20.7	19.9	17.4	14.0	14.2	14.8	14.5	13.7	16.7	16.0	
		MCDBR	30.6	31.0	30.9	27.6	25.3	20.9	18.8	19.4	20.7	21.7	25.0	26.8	
		MCWBR	14.4	12.5	10.9	8.3	7.9	6.9	5.7	6.0	6.4	8.0	10.6	13.3	
	5% WB	MCDBR	28.2	28.4	26.4	23.1	20.4	16.3	15.5	16.3	17.5	18.6	21.7	24.2	
		MCWBR	14.4	12.7	10.6	8.3	8.0	6.4	5.4	5.8	6.1	7.8	10.0	12.8	
Clear-Sky Solar Irradiance	taub	0.367	0.383	0.496	0.511	0.470	0.455	0.456	0.477	0.461	0.463	0.419	0.385		
	taud	2.209	2.181	1.867	1.886	2.075	2.185	2.180	2.096	2.114	2.026	2.067	2.141		
	Ebn at Noon	274	280	254	254	264	266	266	260	260	250	253	261		
	Edh at Noon	39	43	63	64	53	47	47	51	49	50	45	40		
All-Sky Solar Radiation	RadAvg	894	1195	1394	1570	1508	1293	1482	1387	1149	899	939	788		
	RadStd	138	194	128	178	135	225	131	160	166	126	119	117		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.83	N/A	N/A	+1.57	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	+0.83	N/A	N/A	+1.46	N/A	N/A	N/A	-166	+257	+97

Nomenclature: See separate page